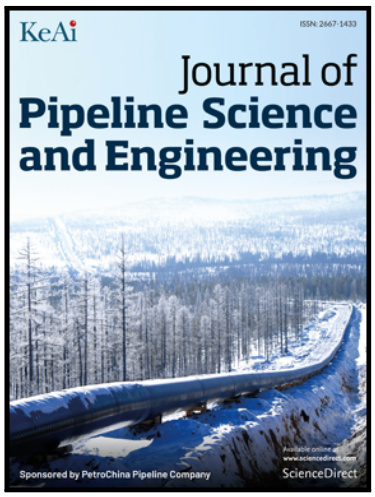
Journal Pre-proof



Research on LLM-based assessment methods for international standards' intelligent contribution to oil and gas pipelines

Bing Liu , Zhonglin Li , Qingquan Duan , Bingchuan Yan , Xiao Tan

PII: S2667-1433(25)00152-0
DOI: https://doi.org/10.1016/j.jpse.2025.100405
Reference: JPSE 100405

To appear in: *Journal of Pipeline Science and Engineering*

Received date: 26 August 2025
Revised date: 23 October 2025
Accepted date: 24 November 2025

Please cite this article as: Bing Liu , Zhonglin Li , Qingquan Duan , Bingchuan Yan , Xiao Tan , Research on LLM-based assessment methods for international standards' intelligent contribution to oil and gas pipelines, *Journal of Pipeline Science and Engineering* (2025), doi: https://doi.org/10.1016/j.jpse.2025.100405

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2025 The Authors. Publishing Services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd.
This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)

Highlights

- **Innovation point name**
- A Method for Evaluating Intelligent Contribution to International Oil and Gas Pipeline Standards Based on Large Language Models
- **Innovation point content**
- First, different large language models are employed to establish an intelligent contribution evaluation index system for international oil and gas pipeline standards.
- Second, the SBERT natural language processing large language model is used to quantitatively extract core indicators. Subsequently, the BWM-entropy weight-game theory combined weighting method is applied to assign weights to the evaluation indicators.
- Finally, the cloud model comprehensive evaluation method is utilized to quantitatively assess China's intelligent contribution to the typical international oil and gas pipeline standard ISO 22974. The evaluation results demonstrate that China has made significant contributions during the development process of the ISO 22974 international standard for oil and gas pipelines, positioning it as a key contributor to ISO 22974.
- Furthermore, leveraging a large language model to assess the intellectual contributions of other relevant stakeholders shows that the Netherlands scored 12.6105, Italy scored 5.4045, and the Russian Federation scored 2.7023, among others. These scores indicate the differentiated professional value exhibited by various countries in the formulation of ISO 22974 according to their respective strengths and needs. Such contributions have injected diverse and critical intellectual input, fostering the global recognition and adoption of the standard.

Journal Pre-proof

Journal Pre-proof

Research on LLM-based assessment methods for international standards' intelligent contribution to oil and gas pipelines

Bing Liu[a] , Zhonglin Li[b] , Qingquan Duan[b] , Bingchuan Yan[c] , Xiao Tan[a]

a.Science and Technology Research Institute, China Oil&Gas Pipeline Network Corporation, Tianjin, China

b.College of Safety and Ocean Engineering, China University of Petroleum (Beijing), Beijing, China

c.National Pipeline Network Group Storage and Transportation Technology Development Co., Ltd, Tianjin, China

ABSTRACT

With economic globalization and the restructuring of the global energy industry, international oil and gas pipeline standards have become a key driver shaping the sector's competitive landscape. Competition for standard-setting authority has thus evolved into a major arena of technological and economic rivalry. Yet, assessing a nation's intelligent contribution to such standards remains challenging. This study proposes an LLM-driven evaluation framework that integrates historical standard development data with advanced natural language processing. A fine-tuned SBERT model, outperforming its non-fine-tuned counterpart in extracting complex indicator data, was employed for quantitative assessment. The methodology combines the Best–Worst Method, entropy weighting, game-theoretic weighting, and a normal cloud model to comprehensively evaluate contributions. ISO 22974 Petroleum and natural gas industries—Pipeline transportation systems—Specification for pipeline integrity assessment served as a case study. Results show that China achieved the highest intelligent contribution score (74.7790), followed by the Netherlands (12.6105), Italy (5.4045), and Russia (2.7023), confirming China as the primary developer of ISO 22974. The findings highlight the strong data processing and learning capabilities of large language models in standardization contexts. The proposed LLM-based framework, integrating AI with conventional evaluation methods, offers significant potential for advancing objective assessments of contributions to international standards.

Keywords： large language model; oil and gas pipelines; international standards; intelligent contribution; assessment methods

Corresponding author: Bing Liu,

Telephone: 13303161808

E-mail: lbcat@126.com

Address: Science and Technology Research Institute, China Oil&Gas Pipeline Network Corporation, Tianjin, China

Author:

1. Bing Liu, lbcat@126.com, Science and Technology Research Institute, China Oil&Gas Pipeline Network Corporation, Tianjin, China
2. Zhonglin Li, 18893752712@163.com, College of Safety and Ocean Engineering, China University of Petroleum (Beijing), Beijing, China
3. Qingquan Duan, dqq@cup.edu.cn, College of Safety and Ocean Engineering, China University of Petroleum (Beijing), Beijing, China
4. Bingchuan Yan, yanbc@pipechina.com.cn, National Pipeline Network Group Storage and Transportation Technology Development Co., Ltd, Tianjin, China
5. Xiao Tan, amoundtan@163.com, Science and Technology Research Institute, China Oil&Gas Pipeline Network Corporation, Tianjin, China

Fund Project: This article is supported by the National Key R&D Program of China "Research on International Standardization in the Field of Oil and Gas Energy Storage and Transportation" (Project No.: 2022YFF0609702).

ABSTRACT

With economic globalization and the restructuring of the global energy industry, international oil and gas pipeline standards have become a key driver shaping the sector's competitive landscape. Competition for standard-setting authority has thus evolved into a major arena of technological and economic rivalry. Yet, assessing a nation's wisdom contribution to such standards remains

challenging. This study proposes an LLM-driven evaluation framework that integrates historical standard development data with advanced natural language processing. Based on the full lifecycle data of international oil and gas pipeline standards development, an evaluation indicator system was constructed by integrating multiple large language models. A fine-tuned Sentence-BERT (SBERT) model was subsequently employed to quantify the core metrics within the wisdom contribution evaluation framework. wisdom contribution scores were then calculated using the Best-Worst Method (BWM), the entropy weight method, the game theory weighting method, and the normal cloud model. Finally, this wisdom contribution evaluation approach was applied to ISO 22974 "Petroleum and Natural Gas Industries—Pipeline Transportation Systems—Specification for Pipeline Integrity Assessment" as a case study.Results show that China achieved the highest wisdom contribution score (74.7790), followed by the Netherlands (12.6105), Italy (5.4045), and Russia (2.7023), confirming China as the primary developer of ISO 22974. The findings highlight the strong data processing and learning capabilities of large language models in standardization contexts. The proposed LLM-based framework, integrating AI with conventional evaluation methods, offers significant potential for advancing objective assessments of contributions to international standards.

Keywords： large language model; oil and gas pipelines; international standards; wisdom contribution; assessment methods

1. Introduction

Oil and natural gas are the lifeblood of modern industrial society, and oil and gas pipelines are critical infrastructure for energy transportation (Li et al., 2022; Chen, 2025). The formulation and improvement of relevant international standards are crucial for ensuring energy security, promoting international trade, and advancing sustainable industry development (Vosta, 2023). These international standards for oil and gas pipelines cover design, construction, operation, maintenance, and other aspects, serving as the basis for harmonizing technical specifications across countries and ensuring safe and reliable operation of pipeline systems (Amaechi et al., 2022; Iqbal et al., 2017; Selvik et al., 2022; Wilkowski et al., 2000; Xia et al., 2023).

However, with structural adjustments in the global energy industry and the emergence of new technologies, traditional methods for evaluating international pipeline standards face unprecedented challenges, especially in assessing stakeholders' wisdom contributions during the standard development process (Mangion, 2012; OECD and International Organization for Standardization, 2016).

The wisdom contribution of an international pipeline standard is defined as the contribution rate of the knowledge or wisdom provided by leading or participating parties during standard development or revision, encompassing overall participation, content inputs, textual contributions, meeting organization, and other aspects.

Currently, there is no established method for evaluating wisdom contributions to international oil and gas pipeline standards. Existing research on pipeline standards mainly focuses on standard internationalization and clean energy transition. For example, [Anonymous, (2020)] proposed a "Belt and Road" oil and gas pipeline standardization cooperation strategy, providing important support for standardization collaboration among Belt and Road countries. Cao et al. (2023) analyzed the status, bottlenecks, and breakthrough paths in the internationalization of China's pipeline standards.

On the application side, the ASME B31.12 hydrogen pipeline standard has established a complete technical specification system for gaseous hydrogen pipelines, laying the foundation for international hydrogen pipeline standardization (ASME, 2023). The U.S. PHMSA issued new regulations in January 2025 that include standards for the design, installation, operation, and maintenance of gaseous carbon dioxide pipelines, further enhancing CO_2 pipeline safety. Despite these developments, none address wisdom contribution evaluation during standard formulation (Reierson and Kammer, 2025).

In recent years, artificial wisdom technologies centered on deep learning have achieved breakthroughs. large language model, in particular, exhibit powerful natural language processing capabilities, as well as strong reasoning and generalization (Ekundayo and Ezugwu, 2025). LLM can deeply understand and semantically mine massive textual data, automatically extracting key information and identifying semantic relationships. They perform complex language tasks at levels approaching or even surpassing human performance. (Chen et al., 2025; Gao et al., 2025; Myers et al., 2024; Youssef et al., 2025).

These capabilities provide a new approach for tackling challenges in evaluating international pipeline standards. Introducing LLM techniques into the evaluation of wisdom contributions to pipeline standards is expected to overcome limitations of traditional methods, achieving comprehensive and precise content analysis, uncovering latent knowledge connections and innovations, and offering technical support for standard formulation, revision, and optimization.

Based on these considerations, this paper conducts an in-depth study of LLM-based methods for assessing wisdom contributions to international oil and gas pipeline standards. The goal is to build an wisdom and efficient evaluation model that, through deep parsing and quantitative analysis of international standard texts, accurately assesses each stakeholder's knowledge contribution during the standard development process. This provides a scientific basis and decision support for high-quality development of pipeline standards. The study systematically investigates LLM technical principles, data processing methods, evaluation indicator system construction, and comprehensive assessment methods and applications, with the aim of achieving innovative results that contribute to the energy industry's digital transformation and international collaboration.

2. Construction of the wisdom contribution indicator system

2.1. The role of LLM in identification and quantification of contributing factors

LLM play a key role in identifying and quantifying contribution elements. During the standard development process, LLM's powerful semantic understanding and information extraction capabilities allow them to effectively process the vast data generated, including standard documents, technical proposals, negotiation records, and related data. LLM can construct networks of technical terms and concepts, accurately parse complex technical semantics and multi-level concept relationships, and track the evolution of terminology across versions. For structured information extraction, LLM use prompt-based extraction methods to extract key contribution elements from unstructured text and structure them into a computable indicator system (Shanahan, 2024; Wu et al., 2024; Zhao et al., 2025).

2.2. Construction of an wisdom contribution assessment indicator system for international standards of oil and gas pipelines

Building a scientifically sound indicator system follows four basic principles (Bi et al., 2021).

(1)Systematic Principle: The indicator system should reflect holistic characteristics. The comprehensive evaluation should be better than a simple sum of sub-indicators, forming an organically connected evaluation framework.

(2)Precision Principle: Selected indicators must have strong representational power, accurately reflecting stakeholders' contributions during standard development and avoiding information distortion.

(3)Exclusivity Principle: Indicators at the same level should have clear boundaries, avoiding conceptual overlap or redundancy to ensure each dimension of evaluation is independent.

(4)Comparability Principle: Both qualitative and quantitative indicators should enable valid horizontal (between standards) and vertical (across time or versions) comparisons.

The key stages of the international standard development life cycle include the Preparatory Stage (PWI), Proposal Stage (NP), Working Draft Stage (WD), Committee Draft Stage (CD), Draft International Standard Inquiry Stage (DIS), Final Draft International Standard Approval Stage (FDIS), and Publication Stage (IS). Based on this life-cycle framework, we used different LLM to analyze data at each stage, yielding candidate indicator sets. **Fig. 1** illustrates the process of constructing an evaluation index system for the wisdom contribution of international oil and gas pipeline standards based on large language models.

Raw data from the international standards development process were first fed into various AI large models. Prior to the first iteration, domain experts manually eliminated irrelevant or unsuitable indicators. Two iterative rounds, combining different AI large models, were then conducted to optimize the indicator set. This process ensured that single-model iterations, which may inadequately capture inter-indicator relationships, were complemented by multi-model optimization to fully reflect the contributions of the leading country. In the final expert refinement step, the results of both AI-driven iterations were synthesized to establish a unified wisdom

contribution assessment framework for the leading party in the oil and gas pipeline standardization process in B. Content Contribution: This dimension directly evaluates the technical knowledge and intellectual outputs contributed to the standard text. B1. Overall text proportion: Represents the proportion of the final standard text derived from the initial proposals, directly reflecting the extent to which a party's proposed framework, structure, and core concepts were accepted. B2. Core technology proportion: Goes beyond textual volume to assess the technical "value content." A fine-tuned SBERT model was used to semantically analyze and quantify the contribution ratio to the core technical sections of the standard, precisely measuring the wisdom input into the most technically significant parts. B3. Expert core technology: To address potential limitations of purely text-based analysis, expert assessments were introduced to evaluate the innovativeness, advancement, and practicality of core technical contributions. This indicator adds a critical qualitative dimension to the wisdom contribution assessment, ensuring accurate quantification of high-level expertise and industry insights.

Table 1.

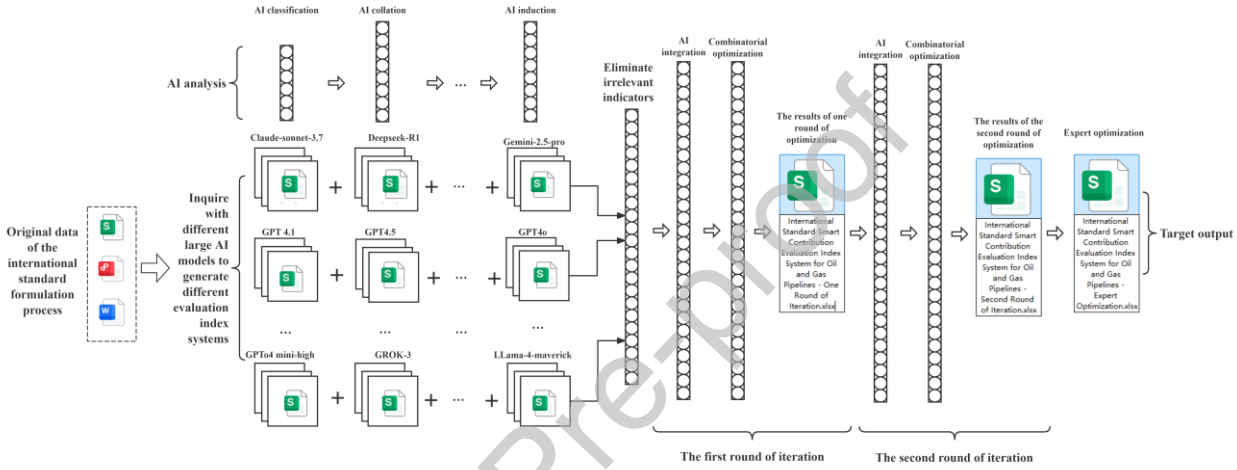


Fig. 1. Construction flowchart of oil and gas pipeline international standards wisdom contribution assessment indicator system

To operationalize the abstract concept of "wisdom contribution," a two-dimensional evaluation system was developed, comprising Organizational Contribution and Content Contribution. Each specific indicator serves as a proxy variable to measure a particular aspect of wisdom contribution, described as follows:

A. Organizational Contribution: This dimension assesses the leadership, coordination, and management efforts necessary to advance the standards development process. A1. Number of proposal approvals: Quantifies the acceptance and influence of submitted proposals. Single-round approvals generally indicate technical soundness, clarity, and broad consensus, reflecting high-quality contributions and industry insight. A2. Number of organized meetings: Reflects leadership and responsibility in agenda setting, guiding technical discussions, and reconciling stakeholder positions. The frequency of meetings organized serves as a key indicator of a party's central driving role in the standardization process. A3. Email response rate: Measures the level of collaborative engagement and sustained attention. High response rates ensure smooth information flow and timely issue resolution, safeguarding the standards development process through effective communication.

B. Content Contribution: This dimension directly evaluates the technical knowledge and intellectual outputs contributed to the standard text. B1. Overall text proportion: Represents the proportion of the final standard text derived from the initial proposals, directly reflecting the extent to which a party's proposed framework, structure, and core concepts were accepted. B2. Core technology proportion: Goes beyond textual volume to assess the technical "value content." A fine-tuned SBERT model was used to semantically analyze and quantify the contribution ratio to the core technical sections of the standard, precisely measuring the wisdom input into the most technically significant parts. B3. Expert core technology: To address potential limitations of purely text-based analysis, expert assessments were introduced to evaluate the innovativeness, advancement, and

practicality of core technical contributions. This indicator adds a critical qualitative dimension to the wisdom contribution assessment, ensuring accurate quantification of high-level expertise and industry insights.

Table 1

Oil and gas pipeline international standards wisdom contribution assessment indicator system table

Primary Indicator	Secondary Indicator	Indicator Description	computing mode
A.Organizational Contribution	A1.Voting pass count	Number of proposals that passed on the first vote.	$y=\begin{cases} 1.0, x = 1, \\ 1.0 - 0.5(x - 1), 3 > x > 1, \\ 0.0, x \geq 3. \end{cases}$ y is the score, and x is the number of approved votes.
	A2.Meetings organized	Number of meetings organized.	$y=\begin{cases} 0\ , x = 0 \text{or } x = 1, \\ 0.2(x - 1)\ , 2 \leq x \leq 5, \\ 1.0\ , x \geq 6. \end{cases}$ y is the score, and x is the number of meetings organized.
	A3.Email response rate	Percentage of emails responded to during the standard development process.	Percentage of emails replied.
B.Content Contribution	B1.Overall text proportion	Proportion of the initial proposal in the final standard text.	Proportion of the initial proposal in the final standard text.
	B2.Core technology	Chapter-by-chapter assessment of whether core technical content was modified.	To distinguish chapters, mainly check whether the core content has been changed, compare the weight of terms and chapters, and finally add up the scores.
	B3.Expert core technology	Expert scoring of core technology contributions.	Expert rating (out of 100)

2.3. LLM-assisted indicator extraction

The general-purpose BERT model is well-known for computing text similarity in NLP. It excels at understanding individual words in context and has good generalization, but it is not specifically designed to capture the meaning of entire sentences or long texts. This can limit its performance in semantic search or finding similar texts (Devlin et al., 2019; Patricoski et al., 2022; Shaik et al., 2024).

Moreover, due to its substantial computational overhead, BERT's efficiency in accurately detecting text similarity is limited. To address these issues, Reimers and Gurevych. (2019) proposed the SBERT model, which optimizes BERT by fine-tuning it on sentence similarity tasks.

Fig. 2 presents the architecture of the SBERT model. Architecture consisting of two identical BERT models with shared weights. Two input sentences are fed into the two identical BERT branches. The outputs of these models (contextual embeddings for each token) are aggregated via a pooling operation (such as mean pooling, max pooling, or using the [CLS] token) to produce fixed-size sentence embedding vectors that represent the entire sentence semantics. This pooling mechanism allows SBERT to focus on the overall sentence representation rather than individual word embeddings.

After pooling, the two sentence embeddings produced by the siamese BERT models are compared using cosine similarity, which measures the closeness of the two vectors in high-dimensional space. During training, a cosine similarity loss function is applied: embeddings of dissimilar sentences are pushed apart (rewarding large distances), while embeddings of similar sentences are pulled closer (penalizing large distances). By iteratively optimizing this loss across the training data, the network weights are adjusted to minimize the loss, effectively learning to encode sentence semantics(Bingi and Yin, 2023; Bölücü et al., 2023; Li et al., 2023).

Journal Pre-proof

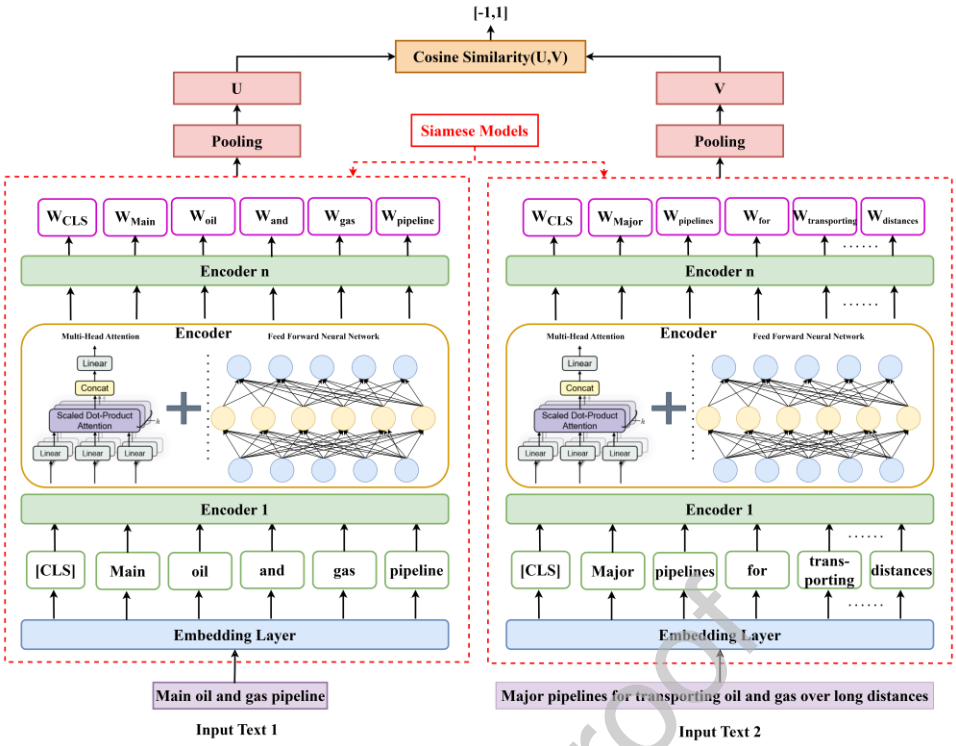


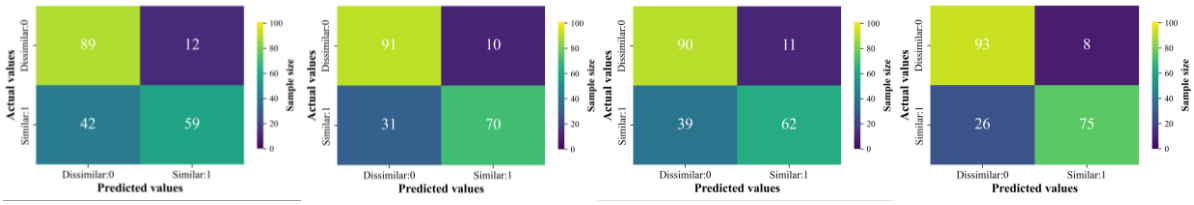
Fig. 2. SBERT model architecture diagram

Because SBERT is trained on general-domain text, it may not compute similarity well in a specific domain. Directly using SBERT to measure similarity of oil and gas pipeline standard texts yields low accuracy and cannot precisely quantify wisdom contributions.

Therefore, we fine-tuned SBERT for the pipeline domain. We used an oil and gas pipeline terminology dataset with definitions. The dataset of oil and gas pipeline terminology and definitions was compiled from authoritative standards such as "Petroleum and Natural Gas Industries-Pipeline Transportation Systems-Terms and Definitions". These standards serve as reliable references in the pipeline industry, containing extensive technical terms and definitions. The dataset comprehensively covers various aspects of pipeline operations including design, construction, maintenance, and safety. The collected terminology accurately reflects the field's characteristics. To enable the model to distinguish between similar terms and definitions, we categorized valid terms and definitions as positive samples, while randomly shuffling corresponding definitions as negative samples. The dataset contains 2020 entries, evenly distributed between positive and negative samples (50% each). We ensured balanced representation of both sample types to maintain data equilibrium. To ensure thorough model training, the dataset is divided into 1,616 training samples, 202 validation samples, and 202 test samples in an 8:1:1 ratio. Similar positive samples are labeled as 1, while dissimilar negative samples are labeled as 0, with a similarity classification threshold set at 0.5. This ensures the fine-tuned model demonstrates both effectiveness and strong generalization capabilities. The cosine similarity loss function is employed for fine-tuning. The process involves encoding two sentence texts into embedding vectors, calculating their cosine similarity, and then comparing the similarity with the true label to compute the loss. When the true label is 1 (similar), the cosine similarity is maximized; when the true label is 0 (dissimilar), the cosine similarity is minimized.

We evaluated SBERT's performance on the validation and test sets before and after fine-tuning. **Fig. 3** depicts the confusion matrices of the model on the validation and test sets before and after fine-tuning. **Table 2** reports Accuracy, Precision, Recall, and F1 Score for the models. The results demonstrate that the fine-tuned SBERT model outperforms the pre-fine-tuned model on all metrics: it better captures the semantics of sentences in the oil and gas pipeline standards domain and accurately measures sentence similarity, showing strong generalization.

Journal Pre-proof



(a) On validation set before SBERT fine-tuning (b) On validation set after SBERT fine-tuning (c) On test set before SBERT fine-tuning (d) On test set after SBERT fine-tuning

Fig. 3. SBERT model evaluation confusion matrices

Table 2

Evaluation metrics for SBERT model on validation and test sets

Dataset	SBERT Model	Accuracy	Precision	Recall	F1 Score
Validation	SBERT(before tuning)	0.7327	0.8310	0.5842	0.6860
Validation	SBERT(after tuning)	0.7970	0.8750	0.6931	0.7735
Test	SBERT(before tuning)	0.7525	0.8493	0.6139	0.7126
Test	SBERT(after tuning)	0.8317	0.9036	0.7426	0.8152

Based on the established indicator system, we extract and quantify data for each indicator. Simple count indicators like “voting pass count”, “meetings organized”, and “email response rate” were extracted manually. The complex indicators “overall text proportion” and “core technology proportion” were quantified using the fine-tuned SBERT model. For the "expert core technology" indicator, a questionnaire was conducted among the expert group members involved in the review of relevant standards during the international standard formulation process of oil and gas pipelines. They were asked to rate the expert strength invested by the leading party in the standard formulation process. Finally, the contribution of expert core technology was obtained by calculating the average value.

3. Comprehensive evaluation method (game theory + cloud model)

To overcome the limitations of single-weighting methods and conduct a scientific, objective, and balanced evaluation of the complex concept of "wisdom contribution," this study employs three weighting methods: the Best-Worst Method (BWM), entropy weighting, and game theory-based combined weighting. The BWM, as a subjective weighting method, effectively reflects expert experience but cannot eliminate subjectivity. The entropy weighting, as an objective method, assigns weights based on data dispersion but may introduce bias from minor data variations. Combining BWM with entropy weighting achieves complementary validation between subjective judgment and objective data. However, simple combinations of subjective and objective weights (e.g., addition or multiplication) may lead to imbalance. Therefore, the game theory-based combined weighting method is introduced to automatically find the optimal balance between subjective and objective weights, ensuring a more stable and balanced final composite weight. Given that wisdom contribution assessment involves multi-level complex indicators, this three-stage combined weighting model meets the scientific and reliability requirements of evaluation, providing a solid foundation for subsequent comprehensive assessments (cloud model).

3.1. Indicator weighting (BWM and entropy)

3.1.1. Best-Worst Method (BWM)

Rezaei (2016) proposed the Best-Worst Method (BWM), it is a multi-criteria decision-making technique to determine indicator weights.

The core concept of BWM is to determine indicator weights through simplified pairwise comparisons. It requires experts to identify the 'best' (most important) and 'worst' (least important) indicators, then compare all other indicators with these two benchmarks. While AHP requires $n(n-1)/2$ pairwise comparisons for all indicators, BWM significantly reduces the number of required comparisons (only $2n-3$), thereby enhancing decision consistency and efficiency. The steps are:

(1)Indicator set definition: Define the evaluation indicator set $X=\{x_1,x_2,...,x_n\}$ with n indicators. Experts choose the best indicator X_B and the worst indicator X_W based on experience.

Journal Pre-proof

(2)Construction of comparison vectors: **Fig. 4** illustrates the process of constructing comparison vectors.Experts (decision-makers) score the criteria using a scale from 1 to 9 to determine the relative importance of each criterion with respect to both the most favorable and the least favorable criteria. Accordingly, two comparison vectors are constructed: $C_B=(C_{B1},C_{B2},...,C_{Bj})$ and $C_W=(C_{1W},C_{2W},...,C_{jW})$. Here, C_{Bj} denotes the relative importance of the most favorable criterion compared to criterion j, where a value of 1 indicates equal importance between C_B and C_{Bj}, and a value of 9 indicates that C_B is extremely more important than C_{Bj}. Similarly, C_{jW} represents the relative unimportance of the least favorable criterion compared to criterion j, where 1 denotes equal unimportance between C_W and C_{jW}, and 9 denotes that C_W is extremely more unimportant than C_{jW}.

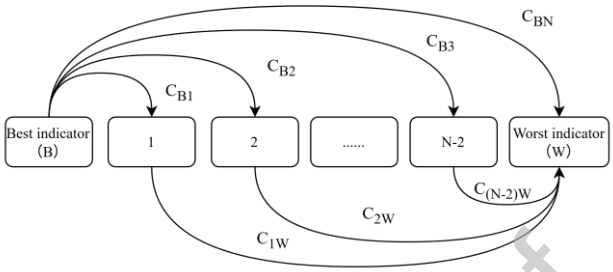


Fig. 4. BWM method comparison process diagram

(3)Determination of optimal criterion weights: A linear programming model is constructed as the objective function to mathematically solve for the optimal weights of the criteria:$\omega_j^* = (\omega_1^*, \omega_2^*, \cdots, \omega_n^*)$;

$$\min \max_j \left\{ \left| \frac{\omega_B}{\omega_j} - a_{Bj} \right|, \left| \frac{\omega_j}{\omega_W} - a_{jW} \right| \right\} \#(1)$$

$$s.t. \begin{cases} \sum_{j=1}^{n} \omega_j = 1 \\ \omega_j \geq 0, j = 1,2,\cdots,n \end{cases} \#(2)$$

Where: ω_B denotes the weight of C_B; C_j represents the criterion vector; ω_j is the weight of C_j, i.e., the actual weight of the criterion; ω_W denotes the weight of C_W; a_{Bj} represents the importance value of C_B relative to C_j; and a_{jW} represents the importance value of C_j relative to C_W. The above equation can be transformed into:

$$\min s.t. \begin{cases} \left| \frac{\omega_B}{\omega_j} - a_{Bj} \right| \leq k \\ \left| \frac{\omega_j}{\omega_W} - a_{jW} \right| \leq k \\ \sum_{j=1}^{n} \omega_j = 1 \\ \omega_j \geq 0 \end{cases} j = 1,2,\cdots,n \#(3)$$

(4)Consistency test: Based on the values obtained from the above equations, the optimal weight sets ω_j^* and k* are determined. The consistency ratio (CR) is calculated using $C_R=\frac{k^*}{C_1}$ (where CR is a predefined threshold). A CR value closer to 0 indicates better consistency, with CR = 0 representing perfect consistency. If m experts participate in the evaluation, the final weights are obtained by computing a weighted average.

$$\omega_j^* = \frac{\sum_{a=1}^{m} \omega_j^{a*}}{m} \#(4)$$

3.1.2. Entropy weight method

For objective weighting, we use the entropy weight method. The steps are:

(1)Data matrix: Construct the initial decision matrix for m samples (standards) and n indicators based on actual data.

Journal Pre-proof

$$X = (x_{ij})_{m\times n} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix} \#(5)$$

(2)Normalization: Standardize the data.

For positive indicators:

$$X'_{ij} = \frac{X_{ij} - \min\left(X_{1j,\ldots,x_{mj}}\right)}{\max\left(X_{1j,\ldots,x_{mj}}\right) - \min\left(X_{1j,\ldots,x_{mj}}\right)} \#(6)$$

For negative indicators:

$$X'_{ij} = \frac{\max\left(X_{1j,\ldots,x_{mj}}\right) - X_{ij}}{\max\left(X_{1j,\ldots,x_{mj}}\right) - \min\left(X_{1j,\ldots,x_{mj}}\right)} \#(7)$$

After processing, a new data matrix X' is obtained:

$$X' = \begin{Bmatrix} x'_{11} & x'_{12} & \ldots & x'_{1n} \\ x'_{21} & x'_{22} & \ldots & x'_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ x'_{m1} & x'_{m2} & \ldots & x'_{mn} \end{Bmatrix}_{m*n} \#(8)$$

(3)Proportion calculation: Calculate the ratio P_{ij} of the value for the j-th evaluation criterion of the i-th sample data:

$$P_{ij} = \frac{x_{ij}^1}{\sum_{i=1}^{m} x_{ij}^1} \#(9)$$

(4)Entropy calculation: Compute the entropy of indicator j:

$$k = \frac{1}{\ln(m)} \#(10)$$

$$e_j = -k * \sum_{i=1}^{m} P_{ij} * \ln\left(P_{ij}\right) \#(11)$$

where k>0 and ej≥0;

(5)Variation coefficient:

$$d_j = 1 - e_j \ (j = 1,2\ldots,n) \#(12)$$

(6)Objective weight: Compute the objective weight

$$w_j = \frac{d_j}{\sum_{j=1}^{n} d_j} \#(13)$$

3.2. Game theory combination weighting

Traditional multiplicative combination of weights can be suboptimal when objective data differences are small. To address this, we use a game theory-based combination weighting method (Rezazadeh et al., 2019).

The steps are:

(1)Weight vectors: Suppose we have L different weighting methods applied to n indicators, yielding L basic weight vectors $q_k=[q_{k1},q_{k2},\ldots,q_{kn}]$, k=1,2,...,L.

(2)Linear combination: Construct a linear combination based on the above L vectors:

$$q = \sum_{k=1}^{L} \gamma_k q_k^T, k = 1,2,\cdots,L \#(14)$$

Where γ_k is the coefficient of the linear combination.

(3)Game optimization: By performing multi-objective game set optimization on the linear combination coefficients of the L vectors, $\gamma=[\gamma_1,\gamma_2,\ldots,\gamma_k]$, the deviation between q and q_k is minimized.

Journal Pre-proof

$$\min \sum_{k=1}^{L} \gamma_k q_k^T - q_k \#(15)$$

Taking the first derivative of the equation yields a system of linear equations:

$$\begin{bmatrix} q_1 \cdot q_1^T & q_1 \cdot q_2^T & \cdots & q_1 \cdot q_L^T \\ q_2 \cdot q_1^T & q_2 \cdot q_2^T & \cdots & q_2 \cdot q_L^T \\ \vdots & \vdots & \ddots & \vdots \\ q_L \cdot q_1^T & q_L \cdot q_2^T & \cdots & q_L \cdot q_L^T \end{bmatrix} \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_L \end{bmatrix} = \begin{bmatrix} q_1 \cdot q_1^T \\ q_2 \cdot q_2^T \\ \vdots \\ q_L \cdot q_L^T \end{bmatrix} \#(16)$$

(4)Normalization: Solve for the optimal linear combination coefficients $\gamma^*=[\gamma_1,\gamma_2,...,\gamma_L]$, and normalize them as follows:

$$\gamma_k^* = \frac{\gamma_k}{\sum_{i=1}^{L} \gamma_k} \#(17)$$

(5)Combined weights: The comprehensive weights of the criteria are thus obtained:

$$w = \sum_{k=1}^{L} \gamma_k^* q_k^T \#(18)$$

3.3. Cloud model-based comprehensive evaluation

The cloud model theory unifies probability and fuzzy logic to reflect the fuzziness and randomness in qualitative human concepts, facilitating conversion between qualitative assessments and quantitative data (Liu et al., 2021; Wang and Duan, 2019; Yao et al., 2019).

A normal cloud model is characterized by three numerical parameters: expectation Ex , entropy En , and hyper-entropy He.

$$\begin{cases} Ex = \dfrac{1}{n}\sum_{i=1}^{n} x_i \\ En = \sqrt{\dfrac{\pi}{2}} \cdot \dfrac{1}{n}\sum_{i=1}^{n} |x_i - Ex| \\ He = \sqrt{\dfrac{1}{n-1}\sum_{i=1}^{n} (x_i - Ex)^2 - En^2} \end{cases} \#(19)$$

Where: x represents the sample observation values; n denotes the number of sample groups; Ex is the expected value of the universe of discourse, representing the central point of cloud droplet distribution within the domain and serving as the ideal mean point for quantifying qualitative concepts; En represents the entropy, indicating the degree of dispersion of cloud droplets within the universe of discourse and characterizing the quantification interval of qualitative concepts, namely the deviation between actual and expected values; He denotes the hyper-entropy of En, describing the degree of aggregation or cohesion of cloud droplets in the cloud model, which corresponds to the thickness of the cloud.

A forward cloud generator uses given cloud parameters to produce “cloud droplets” and corresponding cloud charts, achieving quantitative representation of a qualitative concept. A reverse cloud generator derives the cloud parameters from observed cloud droplets, enabling qualitative interpretation. **Fig. 5** illustrates the cloud generation principle.

Journal Pre-proof

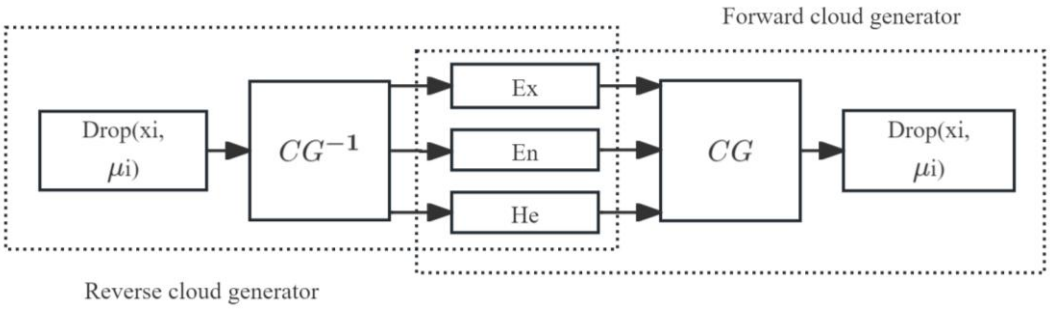


Fig. 5. Cloud generator principle diagram

3.3.1. Evaluation grade clouds

Based on expert consultation and literature, we define five levels for wisdom contribution and their numerical ranges within [0, 100]. These are shown in **Table 3**. The levels are: Low [0.0, 25.0), Rather Low [25.0, 50.0), General [50.0, 75.0), Rather High [75.0, 90.0), and High [90.0, 100.0].

Table 3

Oil and gas pipeline international standards wisdom contribution assessment grade table

Level	Interval	Interpretation
Low	[0.0, 25.0）	Stakeholder made relatively little contribution during standard development and had minimal participation.
Rather Low	[25.0, 50.0）	Stakeholder made some contribution and had a moderate level of participation.
General	[50.0, 75.0）	Stakeholder made notable contributions and was one of the main developers of the standard.
Rather high	[75.0, 90.0）	Stakeholder made significant contributions and was an important developer of the standard.
High	[90.0, 100.0]	Stakeholder made outstanding contributions and played an indispensable role in every stage of standard formulation.

3.3.2. Determination of the digital characteristics of the cloud model

Based on the establishment of assessment level intervals for wisdom contribution evaluation according to international standards for oil and gas pipelines, the standard cloud digital characteristics corresponding to each level interval are calculated. The standard cloud digital characteristic values corresponding to the i-th evaluation interval $[x_i^{min}, x_i^{max}]$ are (Ex_i, En_i, He), where the calculation formulas are:

$$\begin{cases} \mathrm{E}x_\mathrm{i} = \left(x_\mathrm{i}^{\max} + x_\mathrm{i}^{\min}\right)/2 \\ \mathrm{E}n_\mathrm{i} = \left(x_\mathrm{i}^{\max} - x_\mathrm{i}^{\min}\right)/6 \\ \mathrm{He} = \mathrm{k} \end{cases} \#(20)$$

Where k is a constant that needs to be adjusted according to the actual required degree of fuzziness for the project. In this study, a value of 0.5 is adopted.

The cloud characteristic parameters of the evaluation object's indicator scores are calculated using Eq. (21).

$$\begin{cases} Ex = \dfrac{1}{n}\sum_{i=1}^{n} x_i \\ En = \dfrac{1}{n}\sqrt{\dfrac{2}{n}}\sum_{i=1}^{n}|x_i - Ex| \\ S^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}(x_i - Ex)^2 \\ He = \sqrt{|S^2 - E_n^2|} \end{cases} \#(21)$$

Journal Pre-proof

Where: Ex, En, S^2, and He represent the expectation, entropy, variance, and hyper-entropy of the indicators of the object to be evaluated, respectively; x_i denotes the sample observation values of the evaluation indicators.

3.3.3. Indicator cloud leaping algorithm

The second-level indicator score cloud model of the evaluation object is elevated to form the comprehensive evaluation cloud model through the following specific methodology:

$$\begin{cases} Ex = \sum_{j=1}^{n} \mathrm{Ex}_{2j} \times \omega_j \\ En = \sqrt{\sum_{j=1}^{n} \left(En_{2j}^2 \times \omega_j\right)} \\ He = \sum_{j=1}^{n} \left(He_{2j} \times \omega_j\right) \end{cases} \#(22)$$

Where: Ex_{2j}, En_{2j}, and He_{2j} represent the expectation, entropy, and hyper-entropy of the j-th second-level indicator, respectively.

3.4. Comprehensive evaluation method workflow

The core principle of cloud model evaluation is to represent both the actual values of evaluation indicators and assessment levels using cloud models, then obtain the final comprehensive evaluation result cloud through cloud operations, followed by interpretation of the result cloud. **Fig. 6** presents the construction process of the evaluation model for wisdom contribution to international oil and gas pipeline standards, which is developed based on the Lagrange multiplier method for combined weighting and the cloud model. The process mainly involves the following steps:

(1) Data and preprocessing. Select data related to the international standard development process for oil and gas pipelines, and conduct preprocessing of the acquired data.

(2) Establish the evaluation indicator system. Following the construction principles of evaluation indicator systems, utilize LLM to preliminarily construct the evaluation indicator system. Based on comprehensiveness and scientific principles combined with expert assessment, develop the evaluation indicator system for wisdom contribution to international oil and gas pipeline standards, determining 2 first-level indicators and 6 second-level indicators.

(3) Establish the comprehensive evaluation model. First, determine the subjective weights of the evaluation indicator system using the BWM method; second, determine the objective weights of the evaluation indicator system using the entropy weight method, then combine subjective and objective weights using the game theory combined weighting method to obtain comprehensive weights; finally, use the cloud model to obtain comprehensive evaluation results.

(4) Empirical analysis. A specific international oil and gas pipeline standard was selected, and data for the relevant secondary indicators were collected to calculate both subjective and objective weights. The game theory-based combination weighting method was then applied to integrate these weights, yielding the comprehensive weight for each secondary indicator. Subsequently, domain experts provided scores to determine the membership degrees and classify the grades of the secondary indicators. Finally, the cloud model was employed to combine the comprehensive weights with the membership degrees, thereby obtaining the wisdom contribution score for the selected international oil and gas pipeline standard.

Journal Pre-proof

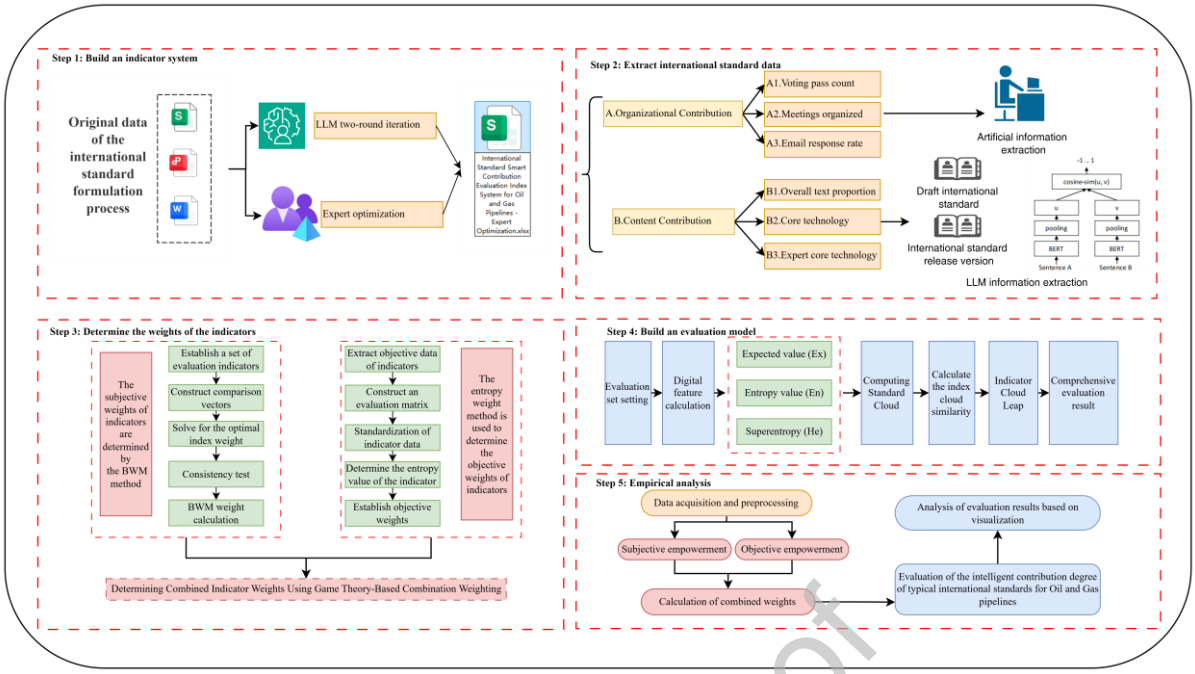


Fig. 6. Comprehensive assessment flowchart of oil and gas pipeline international standards wisdom contribution

4. Empirical analysis results

Considering the characteristics of the oil and gas storage and transportation industry, a representative international pipeline standard is selected for evaluation. ISO 22974, titled Petroleum and natural gas industry—Pipeline transportation systems—Pipeline integrity assessment specification, is an international standard led by China. Officially released in July 2023, it specifies technical requirements and recommendations for pipeline integrity assessment in the petroleum and natural gas sector. ISO 22974 is taken as the research target for assessing wisdom contribution.

4.1. Data acquisition and preprocessing

Data from the development process of ISO 22974 are collected, including draft standards, meeting minutes, meeting videos and audio recordings, voting records, and feedback from participating countries. Data are gathered from the proposal, draft, consultation, and release stages. For quantitative indicators “A1: Number of votes passed,” “A2: Number of meetings organized,” and “A3: Email responses,” numerical quantification is performed. For proportional indicators “B1: Overall text proportion” and “B2: Core technology proportion,” quantification is carried out using a fine-tuned SBERT model. For “B3: Expert core technology,” experts in the relevant field provide scoring for quantification. In total, six quantitative indicators for the international oil and gas pipeline standard are collected in **Table 4**.

Table 4

Different oil and gas pipeline international standards indicator data table

Standard	A1:Vote Pass Count	A2:Meetings Organized	A3:Email Response Rate	B1:Overall Text Proportion	B2:Core Technology Proportion	B3:Expert Core Tech Score
ISO 7055	1	2	0.9542	0.9348	0.4716	57
ISO 10903	1	5	0.9412	0.8711	0.5823	59
ISO 11626	1	6	0.9453	0.9592	0.5964	58
ISO 22974	1	6	0.9524	0.8817	0.6623	57

ISO 24489	1	5	0.9426	0.8378	0.9664	83
ISO 24376	1	6	0.9578	0.8254	0.9662	78

4.2. Comprehensive evaluation of wisdom contribution using the normal cloud model

4.2.1. Determination of indicator weights

To determine subjective weights, six experts in standardization are invited to score the evaluation indicator system for wisdom contribution. The BWM is applied, and Eqs. (1)–(3) are used to compute the weights assigned by each expert. All expert scores pass the consistency test, confirming reliability. The mean of the six experts' weights, obtained using Eq (4), is taken as the subjective weights of the six secondary indicators.The entropy weight method is then applied, using Eqs. (5)–(13), to determine objective weights. Finally, the game-theoretic combination weighting method is used, via Eqs. (14)–(18), to integrate subjective and objective weights, yielding the comprehensive weights in **Table 5**. These weights form the basis for applying the normal cloud model to assess wisdom contribution.

Table 5

Oil and gas pipeline international standards wisdom contribution assessment indicator weight details table

Primary Indicator	Subjective Weight	Secondary Indicator	Subjective Weight	Objective Weight	Combined Weight
A.Organizational Contribution	0.25	A1:Voting pass count	0.0836	0	0.0418
		A2:Meetings organized	0.1258	0.0856	0.1057
		A3:Email response rate	0.0406	0.1802	0.1104
B.Content Contribution	0.75	B1:Overall text proportion	0.1875	0.1746	0.1810
		B2:Core technology proportion	0.3225	0.1667	0.2446
		B3:Expert core technology	0.2400	0.3929	0.3164

4.2.2. Determination of the evaluation set and standard evaluation cloud

Based on data collection and literature review, the evaluation scale for wisdom contribution is defined into five grades:The parameter K is set to the typical value of 0.5. Using the determined grade intervals and Eq. (20), the standard cloud parameters are calculated in **Table 6**. **Fig. 7** illustrates the implementation of a forward cloud generator model in MATLAB, which is employed to generate the standard evaluation cloud chart.

Table 6

Assessment grade and standard cloud parameter table

Level	Interval	Expectation(Ex)	Entropy (En)	Hyper-entropy (He)
Low	[0.0, 25.0）	12.5	4.1667	0.5
Rather Low	[25.0, 50.0）	37.5	4.1667	0.5
General	[50.0, 75.0）	62.5	4.1667	0.5
Rather high	[75.0, 90.0）	82.5	2.5000	0.5
High	[90.0, 100.0]	95.0	1.6667	0.5

Journal Pre-proof

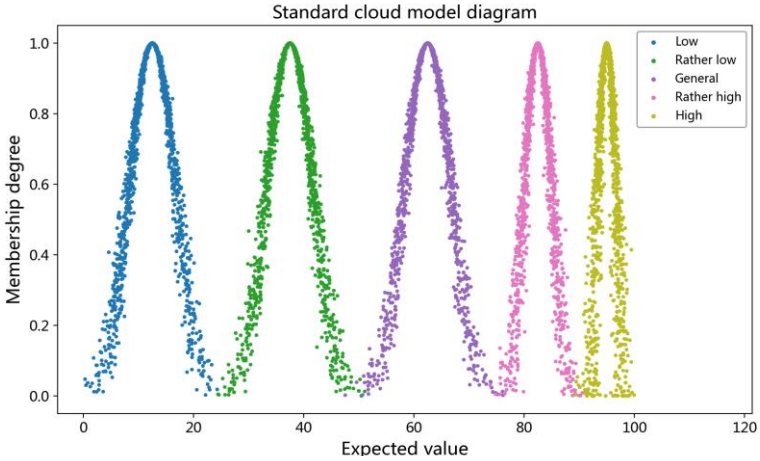


Fig. 7. Standard cloud diagram

4.2.3. Determination of indicator clouds

Based on the objective data of ISO 22974 indicators, the scoring calculation is performed using Eq. (21), and the cloud numerical characteristics of each evaluation indicator are presented in **Table 7**.

Table 7

Assessment indicator cloud digital feature table

Indicator	Expectation (Ex)	Entropy (En)	Hyper-entropy (He)
A1	95.6667	2.9244	0.3385
A2	96.1667	2.7155	1.2620
A3	94.3333	2.0889	0.5508
B1	88.1700	0.6424	0.2182
B2	66.2300	0.6532	0.1010
B3	57.0000	3.7599	2.0647

4.2.4. Calculation of indicator cloud similarity

Eq. (23) is employed to calculate the similarity between each indicator cloud and the standard cloud, while **Fig. 8** illustrates the similarity matrix generated for the secondary evaluation indicators. This facilitates clearer understanding of each indicator's contribution to the overall evaluation. **Fig. 9** presents the indicator cloud chart obtained by inputting the evaluation indicator cloud parameters into the forward cloud generator. These diagrams illustrate the comparison between the known standard cloud digital characteristics and the inverse cloud digital characteristics of the comprehensive evaluation indicators.

$$V(C_1, C_2) = e^{-\sqrt{(Ex_1-Ex_2)^2+(En_1-En_2)^2+(He_1-He_2)^2}} \#(23)$$

Where C_1 and C_2 denote the digital characteristics of the target cloud model and reference cloud model, respectively, expressed as (Ex_1, En_1, He_1) and (Ex_2, En_2, He_2).

The six secondary indicators in the ISO 22974 evaluation exhibit varying levels of contribution. China's contributions are at a high level for voting, meeting organization, email responses, and overall text proportion, but only average for core technology and expert core technology, suggesting the need for greater emphasis on these two aspects in future standard development.

Journal Pre-proof

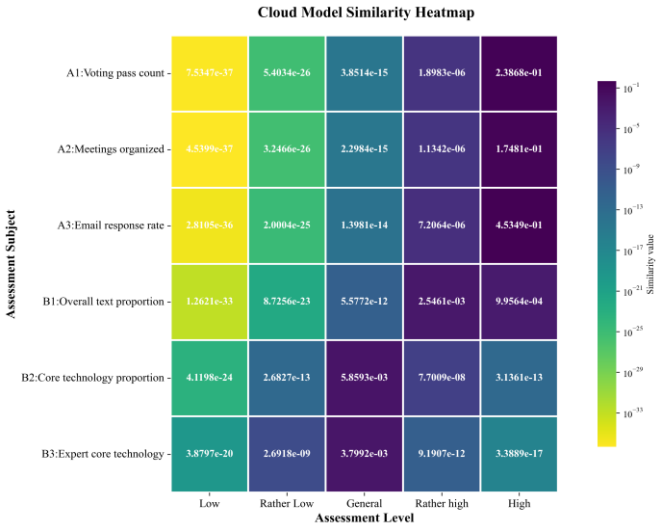
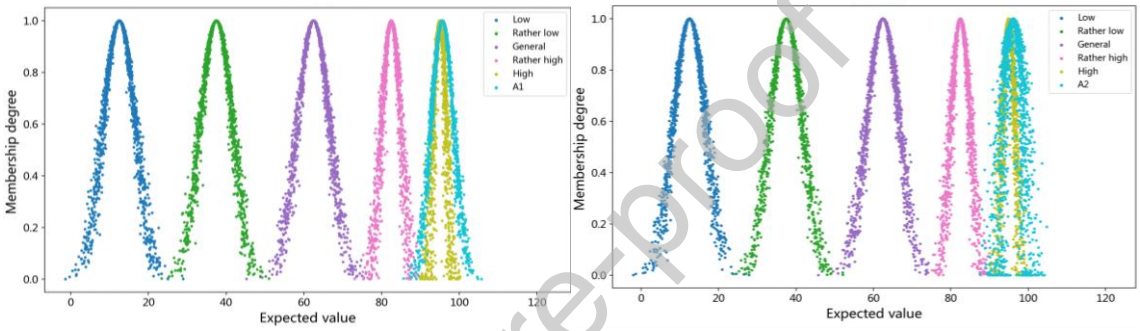
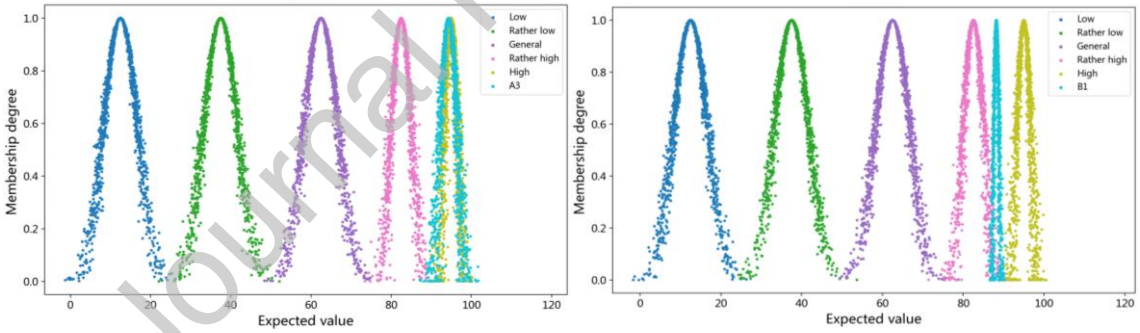


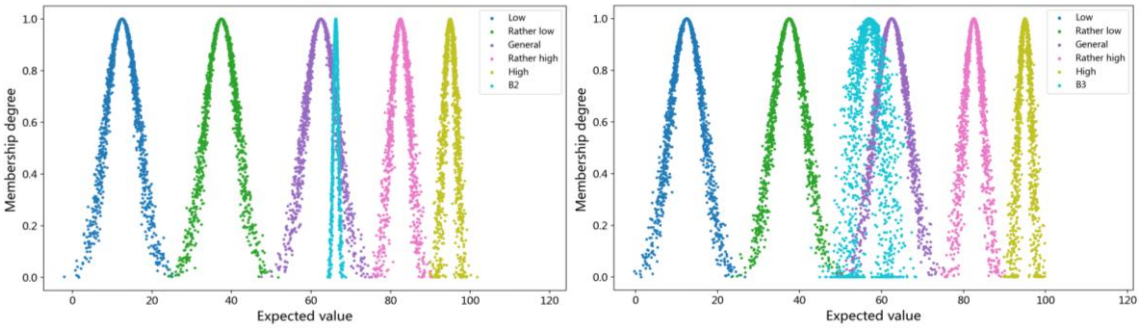
Fig. 8. Assessment indicator cloud similarity heat map



(a) Voting pass count (b) Meetings organized



(c) Email response rate (d) Overall text proportion



(e) Core technology (f)Expert core technology

Fig. 9. Assessment indicator cloud diagram

4.2.5. Comprehensive evaluation results

Journal Pre-proof

Based on the comprehensive evaluation cloud digital characteristics of the secondary indicators and using the indicator cloud aggregation algorithm in Eq. (22), the overall cloud digital characteristics are computed as (74.7790,2.5043,0.9259). This indicates an wisdom contribution score of 74.7790 for China in the ISO 22974 development process.

To further validate the effectiveness and reliability of the evaluation results, the similarity between the comprehensive evaluation cloud and the standard cloud is calculated using Eq. (23), as presented in **Table 8**. The highest similarity is observed between the comprehensive evaluation cloud and the "relatively high" standard cloud. **Fig. 10** illustrates the comprehensive evaluation cloud chart of ISO 22974 generated using the forward cloud generator. The cyan cloud is located between the purple and red clouds, closer to the red, indicating that the comprehensive evaluation result for ISO 22974 is "relatively high." This confirms that China made a significant contribution and served as the leading developer of ISO 22974.

Table 8

Comprehensive assessment cloud diagram similarity calculation results table

Level	Interval	Expectation(Ex_2)	Entropy (En_2)	Hyper-entropy (He_2)	Similarity	Remark
Low	[0.0, 25.0）	12.5	4.1667	0.5	9.7560×10^{-28}	—
Rather low	[25.0, 50.0）	37.5	4.1667	0.5	6.2057×10^{-17}	—
General	[50.0, 75.0）	62.5	4.1667	0.5	4.1254×10^{-6}	—
Rather high	[75.0, 90.0）	82.5	2.5000	0.5	4.3824×10^{-4}	Maximum
High	[90.0, 100.0]	95.0	1.6667	0.5	1.6168×10^{-9}	—

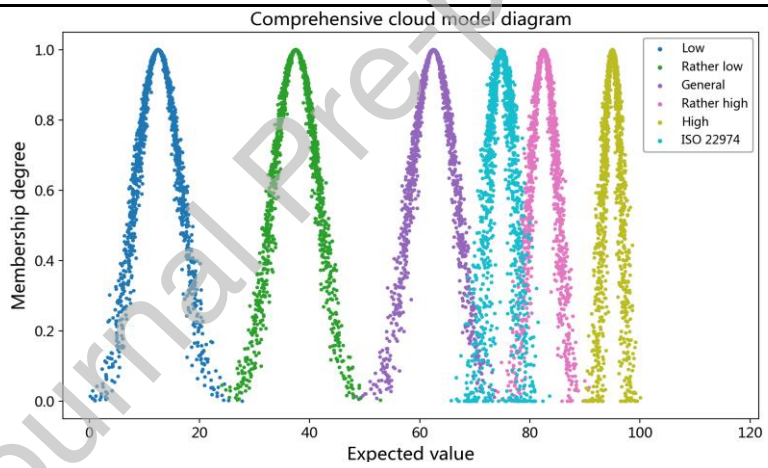


Fig. 10. Comprehensive assessment cloud diagram

4.2.6. Evaluation of the intellectual contributions of other relevant stakeholders

For the evaluation of the wisdom contribution of stakeholders other than China, the assessment was based on the extent to which their proposed amendments were adopted during the development of ISO 22974. A large language model was employed to extract the accepted comments submitted by relevant parties in the formulation process, thereby quantifying their wisdom contributions. **Fig. 11** presents the quantified results of the intellectual contributions of other relevant stakeholders. The results indicate that, among the other stakeholders, the top three countries in terms of wisdom contribution were the Netherlands, Italy, and the Russian Federation. Specifically, the Netherlands ranked first with a remarkable score of 12.6105, demonstrating its substantial involvement and professional influence in the development of ISO 22974. The amendments proposed by the Netherlands not only had a quantitative advantage but also received broad recognition for their quality. These adopted proposals may have involved optimizing the core framework of the standard, refining critical technical specifications, and other significant aspects, thereby further enhancing the scientific rigor and practical applicability of ISO 22974. Italy secured the second position with a score of 5.4045, which, although lower than that of the Netherlands, still reflects a noteworthy contribution. The Russian Federation ranked third with a score of 2.7023, marking another

Journal Pre-proof

contribution of importance. Overall, the other relevant stakeholders provided multi-level support in the formulation of ISO 22974, collectively contributing to improvements in content refinement and expansion of its scope of application. This also reflects the differentiated professional value demonstrated by various countries in international standard-setting processes, shaped by their respective strengths and needs, which injected diverse and pivotal intellectual input into ensuring the global recognition and adoption of the standard.

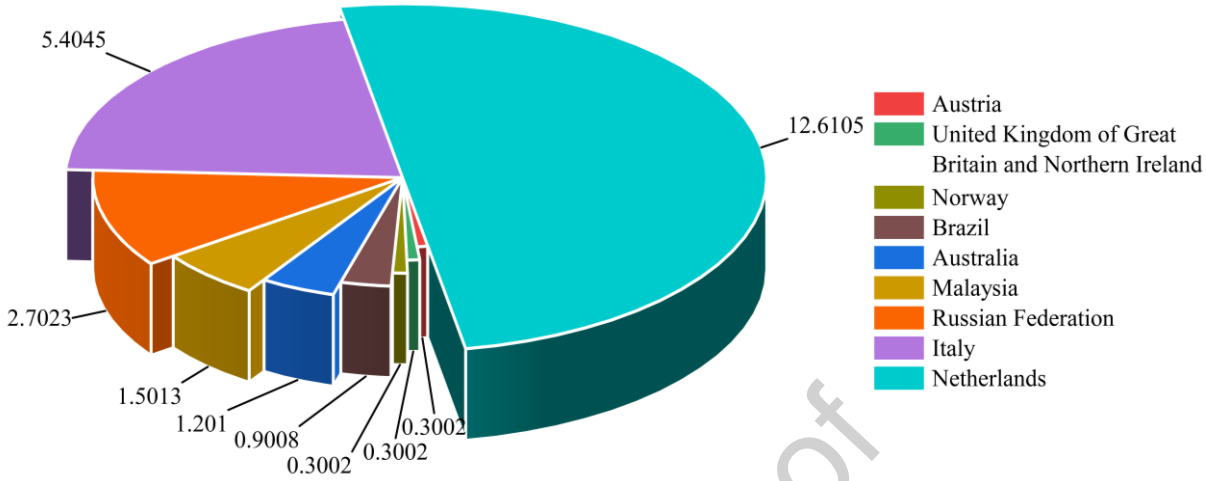


Fig. 11. Intellectual contribution scores of other relevant stakeholders

5. Conclusions

Addressing the research gap in evaluating wisdom contributions to international oil and gas pipeline standards, this study proposes an evaluation method based on large language model. To verify its feasibility, the method is applied to the representative case of ISO 22974, integrating multimodal data extraction and analysis. The main conclusions are as follows:

(1) This study proposes an wisdom contribution evaluation method for international oil and gas pipeline standards based on large language models, which successfully fills the research gap in this field.

First, the application of different large language models has demonstrated novel perspectives in constructing evaluation index systems. Second, by fine-tuning SBERT for the specific domain of oil and gas pipelines, the optimized model enables precise quantification of core metrics in the wisdom contribution assessment framework. This approach allows the model to accurately capture textual semantics and measure sentence similarity, effectively extracting complex indicators for wisdom contribution evaluation. Although current research primarily utilizes LLMs for index system development and text data quantification, this framework holds significant potential for integrating AI with traditional evaluation methods (BWM, entropy weighting method, cloud modeling). It lays the foundation for objectively assessing contributions to international standards.

(2) The quantified index data were applied to a comprehensive evaluation method combining game theory portfolio weighting and cloud model to evaluate the case standard ISO 22974.

The evaluation results show that China's wisdom contribution score in the ISO 22974 development process is 74.7790. By comprehensively assessing the similarity between cloud and standard clouds (with the highest similarity corresponding to the "high" level), the robustness of the evaluation model is strongly verified, and it is confirmed that China played an important and leading role in the formulation of the ISO 22974 standard. Combining China's position in the oil and gas pipeline industry and its standardization strategy, the deeper reasons for its leading position are analyzed as follows:

Journal Pre-proof

a. Policy aspects: China has elevated "standardization" to the level of a national strategy, and actively participating in the formulation of international standards is an important part of the country's efforts to enhance its global governance discourse power and "going global" strategy.

b. In terms of core technologies: China possesses one of the world's longest and most complex long-distance oil and gas pipeline systems, accumulating extensive experience in the design, construction, operation, and integrity management of large-scale and high-difficulty pipelines. These rich experiences have contributed to the main framework of international oil and gas pipeline standards and a large number of foundational texts, making China's technical concepts in oil and gas pipelines the core of international standards and laying the technical foundation for them.

c. Organizational management: The efficient organizational management ensures that ISO 22974 can successfully complete the review, consultation and release of each stage within the scheduled international standard development cycle.

(3) The evaluation results of the intellectual contributions from other stakeholders of ISO 22974 highlight the differentiated professional values demonstrated by countries in standard development based on their respective strengths and needs. The evaluation results show that, apart from China, the Netherlands (12.6105), Italy (5.4045), and Russia (2.7023) are the three countries with the highest contributions. The Netherlands achieved the highest contribution score, reflecting its substantial participation and professional influence in standard development. This demonstrates that multi-stakeholder participation has provided diverse and critical knowledge inputs for the refinement of ISO 22974's content and expansion of its scope of application, ultimately promoting the global recognition and adoption of the standard. This will more clearly highlight the importance and value of this study in terms of methodological innovation and practical application.

(4)While the evaluation framework proposed in this study fills an academic gap, its greater value lies in providing practical guidance for real-world applications. The framework's application scenarios include:

a. National level: It can assist in the top-level design of national standardization strategies, optimize resource allocation through quantitative evaluation, and provide data-driven decision support for negotiations on international standard-setting dominance. Based on evaluation results, a country can identify its strengths and weaknesses in "content contribution" and "organizational contribution," thereby formulating differentiated participation strategies.

b. Enterprise-level: This framework can serve as a tool for companies to measure the global impact of their technologies. By enhancing their 'wisdom contribution' in international standards, enterprises can accelerate global market access for their technologies and products. Moreover, it provides a fresh perspective for analyzing competitors' standardization strategies.

c. At the international organization level: As a management tool to enhance the transparency, balance, and efficiency of standard-setting processes. International standard organizations can use this framework to assess the health of projects, ensure broad stakeholder participation and contributions, and thereby enhance the global credibility and influence of the final published standards.

References

Journal Pre-proof

Amaechi, C.V., Hosie, G., Reda, A., 2022. Review on Subsea Pipeline Integrity Management: An Operator's Perspective. E nergies 16, 98. https://doi.org/10.3390/en16010098.

Bi, Q.-P., Li, Y.-C., Shen, C., 2021. Screening of Evaluation Index and Construction of Evaluation Index System for Mine Ventilation System. Sustainability 13, 11810. https://doi.org/10.3390/su132111810.

[Anonymous 2007] Details omitted for double-anonymized reviewing.

Bingi, Y., Yin, Y., 2023. An Analysis of Embedding Layers and Similarity Scores using Siamese Neural Networks. https:// doi.org/10.48550/arXiv.2401.00582.

Bölücü, N., Can, B., Artuner, H., 2023. A Siamese Neural Network for Learning Semantically-Informed Sentence Embeddi ngs. Expert Systems with Applications 214, 119103. https://doi.org/10.1016/j.eswa.2022.119103.

Cao, Y., Tan, X., Liu, B.,Chang, X.R., Zheng, S.L., 2023. Research on the status,bottleneck and breakthrough path of the internationalization of Chinese oil and gas pipeline standards. J. China Standardization, (17), 65-70. https://doi.org/10.3 969/j.issn.1002-5944.2023.17.009.

Chen, C., Zhao, K., Leng, J., Liu, C., Fan, J., Zheng, P., 2025. Integrating large language model and digital twins in the context of industry 5.0: Framework, challenges and opportunities. Robotics and Computer-Integrated Manufacturing 94, 102982. https://doi.org/10.1016/j.rcim.2025.102982.

Devlin, J., Chang, M.-W., Lee, K., Toutanova, K., 2019. BERT: Pre-training of Deep Bidirectional Transformers for Langu age Understanding. https://doi.org/10.48550/arXiv.1810.04805.

Ekundayo, O.S., Ezugwu, A.E., 2025. Deep learning: Historical overview from inception to actualization, models, applicatio ns and future trends. Applied Soft Computing 181, 113378. https://doi.org/10.1016/j.asoc.2025.113378.

Gao, M., Hu, X., Yin, X., Ruan, J., Pu, X., Wan, X., 2025. LLM-based NLG Evaluation: Current Status and Challenges 5 1.

ASME, 2023. ASME B31.12-2023: Hydrogen piping and pipelines ASME code for pressure piping, B31. American Society of Mechanical Engineers, New York.

Iqbal, H., Tesfamariam, S., Haider, H., Sadiq, R., 2017. Inspection and maintenance of oil & gas pipelines: a review of p olicies. Structure and Infrastructure Engineering 13, 794–815. https://doi.org/10.1080/15732479.2016.1187632.

Patricoski, J., Kreimeyer, K., Balan, A., Hardart, K., Tao, J., J.H.M.T.B.I., 2022. An Evaluation of Pretrained BERT Model s for Comparing Semantic Similarity Across Unstructured Clinical Trial Texts, in: Mantas, J., Hasman, A., Househ, M. S., Gallos, P., Zoulias, E., Liaskos, J. (Eds.), Studies in Health Technology and Informatics. IOS Press. https://doi.org/ 10.3233/SHTI210848.

Li, R., Cheng, L., Wang, D., Tan, J., 2023. Siamese BERT Architecture Model with attention mechanism for Textual Sem antic Similarity. Multimedia Tools and Applications 82, 46673–46694. https://doi.org/10.1007/s11042-023-15509-4.

Li, Z., Liang, Y., Ni, W., Liao, Q., Xu, N., Li, L., Zheng, J., Zhang, H., 2022. Pipesharing: economic-environmental benef its from transporting biofuels through multiproduct pipelines. Applied Energy 311, 118684. https://doi.org/10.1016/j.apen ergy.2022.118684.

Liu, X., Zhang, M., Sun, Z., Zhang, H., Zhang, Y., 2021. Comprehensive evaluation of loess collapsibility of oil and gas pipeline based on cloud theory. Sci Rep 11, 15422. https://doi.org/10.1038/s41598-021-94882-2.

Mangion, D., 2012. Stakeholders' Influence and Contribution to Social Standards Development: The Case of Multiple Stake holder Approach to ISO 26000 Development. Social and Environmental Accountability Journal 32, 115–115. https://doi. org/10.1080/0969160X.2012.718910.

Myers, D., Mohawesh, R., Chellaboina, V.I., Sathvik, A.L., Venkatesh, P., Ho, Y.-H., Henshaw, H., Alhawawreh, M., Berdi k, D., Jararweh, Y., 2024. Foundation and large language models: fundamentals, challenges, opportunities, and social i mpacts. Cluster Comput 27, 1–26. https://doi.org/10.1007/s10586-023-04203-7.

OECD, International Organization for Standardization, 2016. The Case of the International Organization for Standardization: International Regulatory Co-operation and International Organisations. OECD. https://doi.org/10.1787/bb6b0620-en.

Chen, P.C., 2025. Advancements and future outlook of safety monitoring, inspection and assessment technologies for oil an d gas pipeline networks. Journal of Pipeline Science and Engineering 100267. https://doi.org/10.1016/j.jpse.2025.100267.

Journal Pre-proof

Reierson K, Kammer R. PHMSA Releases Notice of Proposed Rulemaking to Enhance Safety of Carbon Dioxide Pipelines. Better Energy. 2025 Jan 22. Available from: https://betterenergy.org/blog/phmsa-releases-notice-of-proposed-rulemaking-to-enhance-safety-of-carbon-dioxide-pipelines/ [Accessed 21 August 2025].

Reimers, N., Gurevych, I., 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. https://doi.org/10.48550/arXiv.1908.10084.

Rezaei, J., 2016. Best-worst multi-criteria decision-making method: Some properties and a linear model. Omega 64, 126–130. https://doi.org/10.1016/j.omega.2015.12.001.

Rezazadeh, A., Talarico, L., Reniers, G., Cozzani, V., Zhang, L., 2019. Applying game theory for securing oil and gas pipelines against terrorism. Reliability Engineering & System Safety 191, 106140. https://doi.org/10.1016/j.ress.2018.04.021.

Selvik, J.T., Alhanati, F.J.S., Signoret, J.-P., 2022. Evaluation of guidance provided by international standards on metrics and timelines for run-life estimation of oil and gas equipment. Life Cycle Reliab Saf Eng 11, 61–74. https://doi.org/10.1007/s41872-022-00185-4.

Shaik, A., Saxena, S., Gupta, M., Parveen, N., 2024. Heterogeneous data-based information retrieval using a fine-tuned pre-trained BERT language model. Multimedia Tools and Applications 83, 59537–59559. https://doi.org/10.1007/s11042-023-17868-4.

Shanahan, M., 2024. Talking about Large Language Models. Commun. ACM 67, 68–79. https://doi.org/10.1145/3624724.

Vosta, M., 2023. International energy trade: The EU's position and energy security. Politics in Central Europe 19, 817–833. https://doi.org/10.2478/pce-2023-0036.

Wang, X., Duan, Q., 2019. Improved AHP–TOPSIS model for the comprehensive risk evaluation of oil and gas pipelines. Pet. Sci. 16, 1479–1492. https://doi.org/10.1007/s12182-019-00365-5.

Wilkowski, G., Stephens, D., Krishnaswamy, P., Leis, B., Rudland, D., 2000. Progress in development of acceptance criteria for local thinned areas in pipe and piping components. Nuclear Engineering and Design 195, 149–169. https://doi.org/10.1016/S0029-5493(99)00245-9.

Wu, L., Zheng, Z., Qiu, Z., Wang, H., Gu, H., Shen, T., Qin, C., Zhu, C., Zhu, H., Liu, Q., Xiong, H., Chen, E., 2024. A survey on large language models for recommendation. World Wide Web 27, 60. https://doi.org/10.1007/s11280-024-01291-2.

Xia, T., Dai, Z., Huang, Z., Liu, L., Luo, M., Wang, F., Zhang, W., Zhou, D., Zhou, J., 2023. Establishment of Technical Standard Database for Surface Engineering Construction of Oil and Gas Field. Processes 11, 2831. https://doi.org/10.3390/pr11102831.

Yao, X., Deng, H., Zhang, T., Qin, Y., 2019. Multistage fuzzy comprehensive evaluation of landslide hazards based on a cloud model. PLoS ONE 14, e0224312. https://doi.org/10.1371/journal.pone.0224312.

Youssef, L., Elhoussaine, Z., Soufiane, N., Noureddine, M., 2025. Enhancing Arabic aspect category detection using large language models (LLMs). Results in Engineering 26, 105049. https://doi.org/10.1016/j.rineng.2025.105049.

Zhao, W., Yang, X., Lyu, Z., Xu, C., Guan, Z., 2025. Road of Large Language Model: Source, Challenge, and Future Perspectives. Research 8, 0655. https://doi.org/10.34133/research.0655.

Funding Statement

This article is supported by the National Key R&D Program of China "Research on International Standardization in the Field of Oil and Gas Energy Storage and Transportation" (Project No.: 2022YFF0609702).

Journal Pre-proof

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Journal Pre-proof